

Three Little Birds

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Abstract

This report presents the development process of Three Little Birds, an interactive data visualization project focused on bird strike incidents and bird migration patterns. Using large-scale real-world datasets, the project combines multiple linked visualizations, including an interactive world map, aircraft heatmaps, bar charts, and dynamic filtering tools, to explore the geographical and operational impact of bird strikes. The report documents the project's progression from dataset selection and exploratory analysis to prototyping, implementation, and final refinement. Special attention is given to challenges related to data quality, visualization design, interactivity, performance optimization, and layout decisions. Through collaborative brainstorming, iterative design, and continuous peer feedback, we developed a cohesive and engaging visualization platform that supports both analytical exploration and storytelling.

1. PATH TO FINAL RESULT

1.1. The Finding Phase

This project started off by searching for datasets. The search was done individually by each group member. The idea was that each member would look into datasets that they would deem interesting and would be sufficient in terms of quality. After considering different datasets, we decided to use a bird strikes dataset suggested by Samuel. During our discussion, we already brainstormed about visualisation ideas, where the initial idea of the map arose and if we could combine it with a different datasets. Our thought was to combine the bird strikes dataset with one of a bird migration. However, there were a few problems for finding a good bird migration dataset. Most were synthetic, which we did not want to use, others were from literature reviews. We decided on the literature review dataset, as we wanted “real” data, but still left our main focal point on the bird strikes, and made the bird migrations an additional but optional feature.

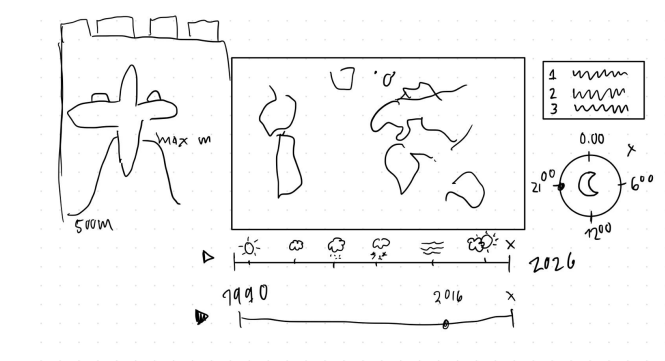


Figure 1. First sketch by Roxane Jaecklin

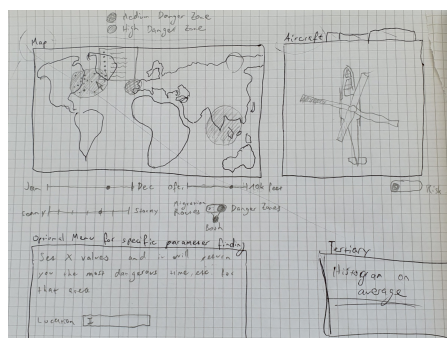


Figure 2. First sketch by Martin Fährnich

1.2. The Exploring Phase

We continued on exploring those two datasets, as M1 required, using a shared jupyter notebook. With these explorations we were able to gain some valuable insights about the consistency and quality of our datasets. This helped in showing what data entries could be used in the final visualisations, as not every column had consistent entries. The created visuals for the exploration can be found on the website of the final visualisation. During this time we met often, to discuss the potential and values of the datasets. It was important for our group that we had a shared understanding of the dataset and our goals for the final visualisation. These discussions were productive as they helped us find our main goals and what we wanted to concentrate on for the journey yet to come. We were able to conclude that our datasets had great potential, which motivated us to move on onto the next step.

1.3. The Brainstorming Phase

The brainstorming phase began with a meeting, where each member made a small sketch on how they imagined the final product would look, without discussing anything beforehand. This was a crucial and necessary step to take, as it enabled us to discuss a lot of new ideas, solidify existing ideas and create a shared consensus where everyone would be happy and motivated to work on the project. These initial sketches are shown in Figures 1, 2 and 3. After we had discussed each sketch in detail we summarised all sketches into one first final draft,

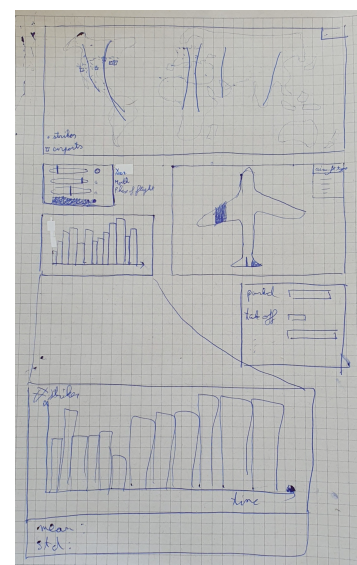


Figure 3. First sketch by Samuel Waridiel

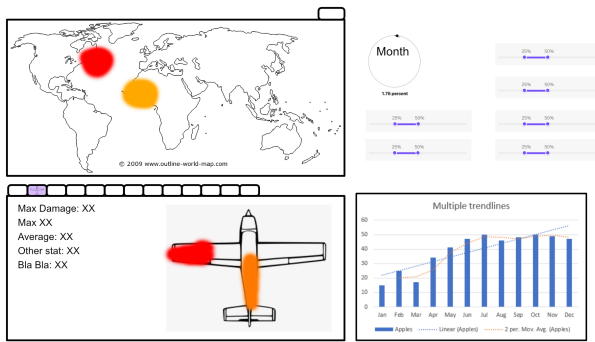


Figure 4. First combined prototype by all three members.

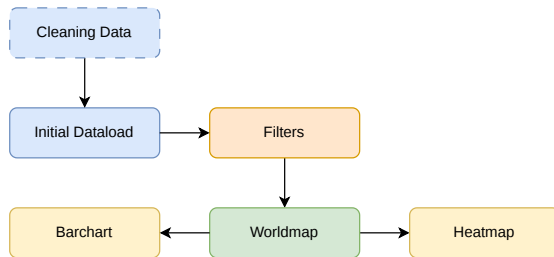


Figure 5. Dataflow of the two datasets through the application.

which is shown in Figure 4 and was also the sketch for M2. These sketches present four main parts of the visualisation:

- An interactive world map, the primary visualisation
- An aircraft heatmap, the secondary visualisation
- A barplot, the third visualisation
- Sliders

There were also other issues, such as setting up a website and creating the main html with the sketched layout. We decided to plan the entire project on Github which included writing issues, assigning them to team members and also assigning those issues to milestones. This helped us to see what was still left to do but also how far other teammates were with their tasks. During this creative time, we also joked about the initial idea of playing the song Danger Zone by Kenny Loggins on the website, as we called the bird strike areas danger zones.

1.4. The Skeleton Phase

This phase was mostly for M2. We were able to finish our tasks two weeks in advance of the deadline, thanks to our in-depth discussion and our organisation on Github. The initial setup of the website was done by Samuel and Martin. Once the website was up and running, we were able to start on our respective tasks. The skeletons were implemented in a short amount of time and each member wrote the report for their part of the visualisation. During this time everyone worked in their corner and communication was mostly via chat. M2 helped to give us an initial idea of how the website might look in the end.

1.5. The Fluid Phase

After submitting the second milestone, the tasks for the group were still pretty clear, since we defined the most important aspects in the Github issues in our repository. Each member continued working on the visualisations they prepared for milestone two. However, first dependencies between the visuals arose which meant another meeting was necessary. We realised that each member loaded the data separately in their script, which made the entire application very inefficient. In addition to fixing this issue we decided to discuss the

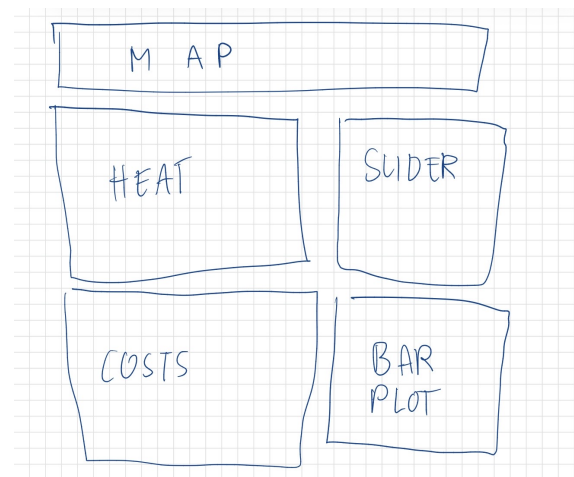


Figure 6. Layout with additional cost heatmap.

general dataflow. This enabled us to finalize the capabilities of each visualisation and how it would influence the other visualisations. The final dataflow can be seen in Figure 5. Before we load the data, unnecessary columns were dropped to decrease the data size. The dataflow starts with loading the data once. The main filtering part is done by the sliders which is why they come before any visualisation. Then follows the map, since it should be possible to implement filters based on danger zones in later. Finally, the filtered data reaches the heatmap and the barplot. The website still took a few seconds to load, which was expected, as the dataset is large. To make the loading a little more eventful, we added the loading icons of planes.

During this time we reopened the discussion for the layout, as we were not happy how our visualisations were displayed. One idea was to make the different sections “drag and droppable”, so that each user would be able to define by themselves on how to arrange the visualisations in the way it would suit them. Ultimately we decided on the layout in Figure 6, as we realized that we wanted to show costs that resulted from birdstrike incidents, also visualized with a heatmap. This extended our visual storytelling by not only presenting the visualisations to the target audience of flight path planners but also of insurance companies, to determine their rates on certain flight routes. This phase was marked by experimenting and rediscussing foundations, but also having fun as it was where the play button for the song Danger Zone was first added.

1.6. The Concretizing Phase

In this final implementation phase, each member was focused on polishing their visualisations. As the visualisations were nearing the final form, we once again had issues with the layout. The biggest one being with the heatmap and the costs. The reason was that it made the entire visualisation more split up, which made it harder to see all visualisations at once without having a big monitor. Being able to see all visualisations is an important feature of our project, as we want to be able to interact with the objects and see at the same time how it influences other components.

This is why the layout was changed again according to Figure 7. As we didn't want to lose the cost-insight, a tooltip was implemented to show all the additional information. The tooltip was very detailed which is why an additional checkboxes-field as seen in Figure 8 was added, so that the shown information could be customized. This section however was quickly removed after a peer review showed how impractical the extra information was. The tooltip remained in a more concise way, only showing a certain amount of detailed information. This cost-story inspired us to add it as an additional dimension in our barplot. Specific costs can be seen when hovering over an aircraft heatmap and also in the barplot.

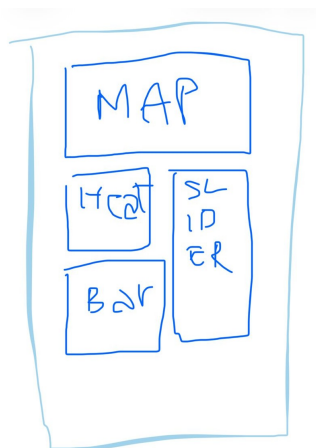


Figure 7. Revised Layout after removing the cost heatmap.

It was during this stage that the interactive and dynamic elements were added to the mostly static map in M2. Thanks to the resources in the OpenLayers package, they were quite simple and straightforward to implement, although some challenges did arise. The most notable difficulty came from the WebGL layer that was being used instead of a regular vector layer to render the points. This choice was made to better handle the large dataset but limited the options in terms of styling and interactions. These difficulties were overcome by adding extra layers to contain the extra information that we wanted to display. The optional migration paths, from the secondary dataset, were also added during this stage.

The website seemed to get close to the goal that we all imagined together, but there were still some issues that had to be addressed. This is why we once again sat together where we critiqued each other's visualisations. Thanks to this we were able to create a list of issues that either had to be added, fixed or changed. The following things were the final tasks:

- Map:
 - Make the clusters transparent
 - Add a number to the clusters to see their actual size
 - Change color scheme from blue to red to yellow to red
- Aircraft Heatmaps:
 - Scale svgs dynamically
 - Implement other and unknown tab
 - Update initial svg of airplane
- Sliders:
 - Allow selection of multiple categorical fields
 - Make them take less space
- Make styling uniform throughout the website

Before we could move onto the final touches of this project, we were once again unhappy with the layout. We still felt that we were not able to see the visualisations as we imagined them. This is why we decided to move the sliders next to the map, so that you can directly see the interaction of the sliders on the map and also see the other two visualisations at the same time. After that the administrative tasks had to be done, such as completing the README, cleaning the code and writing this process book.

1.7. The Creative Phase

In the same meeting where we discussed the final issues, we also decided that we wanted to create our own version of the song Danger Zone for this project. This was the case, because of our running joke and due to copyright issues, but also because we had a lot of fun

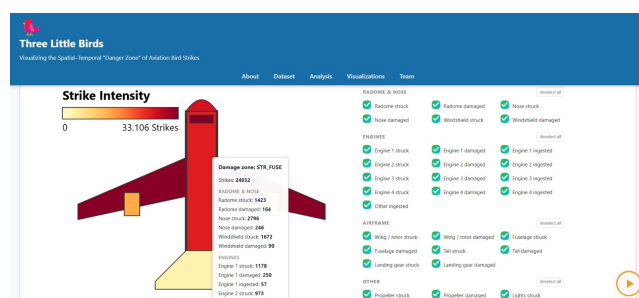


Figure 8. Checkboxes for the tooltip inside the heatmap.

collaborating together and this song would be the cherry on top that ties this project together. To make the song fit even more into the projects vision, we changed the lyrics of “Highway to the Danger Zone” into “Flightpaths through the Danger Zone” and “Ride into the Danger Zone” into “Don’t fly into the Danger Zone”.

2. MAIN CHALLENGES

Although we had pretty smooth sailing throughout the project, the initial phases came with quite a few challenges. The first big challenge we faced was figuring out what kind of visualisations we wanted to create based on our two datasets. We knew that we wanted to implement a map to show bird strikes but we were not sure on how we wanted to combine them with the bird migrations. Having these two huge datasets was very overwhelming for all of us, as we kept jumping onto different ideas and discarding them as quickly as they came. Once we decided to put our focus onto the bird strikes, it made the decision process a bit easier but there were still lots of variables to decide on. This is where we arrived at the next challenge. As the bird strike dataset entails a lot of different measurements and variables we were once again discussing what data we wanted to use and visualise. This is where the exploration of M1 was very helpful in identifying what variables had the most usable data. Sketching the visualisations individually and discussing each member's ideas was the final touch we needed to determine what exactly we wanted to achieve with this project. Finally, the layout was a challenge that followed us through the entire project until the very end.

3. PEER ASSESSMENT

The workload (except for the creative parts) was split approximately equally among the team members and we lived in a work environment of collaboration. Here more detailed information about the contributions of each member:

Done by all members

- Dataset Research
- Writing on the report for M1
- Writing on the report for M2
- Writing on the report for M3

Martin Fähnrich

- Data Exploration of the Bird Strikes dataset
- Cleaning up the Bird Strikes dataset
- Setting up website via Vercel
- Visualisation 2 (Heatmap)
 - Creating tabs for the different aircrafts
 - Creating SVGs for the aircraft types
 - Creating the heatmap for all the aircraft section

Roxane Jaecklin

- Researching related work
- Visualisation 3 (Barchart and Filters)
- Sliders
- Final Layout

Samuel Waridiel

- Visualisation 1 (Worldmap)
- Composing our version of the song Danger Zone
- Data flow